Abstract

EnviroSafe is an AI-based Environmental Compliance Monitoring and Rule-Based Risk Prediction System designed to help industries monitor and manage environmental compliance efficiently. The system is intended for multiple industry types such as textile, chemical, pharmaceutical, cement, food processing, paper, and steel industries, each having different environmental standards. The administrator configures industry-specific compliance standards, while inspectors record environmental parameters such as air quality, water quality, and waste disposal details during inspections. The system automatically compares the recorded values with the predefined standards to determine whether an industry is compliant or non-compliant, thereby reducing manual effort and improving the accuracy of environmental monitoring.

The proposed system uses a rule-based risk prediction approach, where predefined IF–THEN rules classify industries into Low, Medium, or High Risk based on detected compliance violations. It also provides suitable corrective measure recommendations to help industries address environmental issues and improve compliance.

As an affordable IoT enhancement, the system integrates low-cost IoT devices (under ₹1000), including an ESP32 microcontroller, DHT11 temperature and humidity sensor, and MQ-135 air quality sensor. These sensors collect real-time environmental data and transmit it to the backend through Wi-Fi. The collected sensor values are automatically compared with predefined compliance standards, enabling continuous monitoring and more accurate environmental assessment without expensive industrial equipment.

The application will be developed using Angular for the frontend, ASP.NET Core for the backend, and SQL Server as the database. As a scalable mini project, EnviroSafe can be further extended by integrating additional IoT sensors, Machine Learning for intelligent risk prediction, automated notifications, and advanced analytics to support smarter environmental decision-making.